

Custom ISA RISC-V IoT Security Processor

Instruction-Type Waveform Report for Submission

PROJECT

risc_aes_custom_ISA

SIMULATION RESULT

PASS 0x00000053, FAIL 0x00000000

VCD SOURCE

riscv_core_tb.vcd

TOTAL TIME

16550.0 ns

This PDF groups important waveforms by instruction or subsystem type. Each page includes beginner notes explaining what to watch.

1. R-Type Instructions

ADD, SUB, SLL, SLT, SLTU, XOR, SRL, SRA, OR, AND

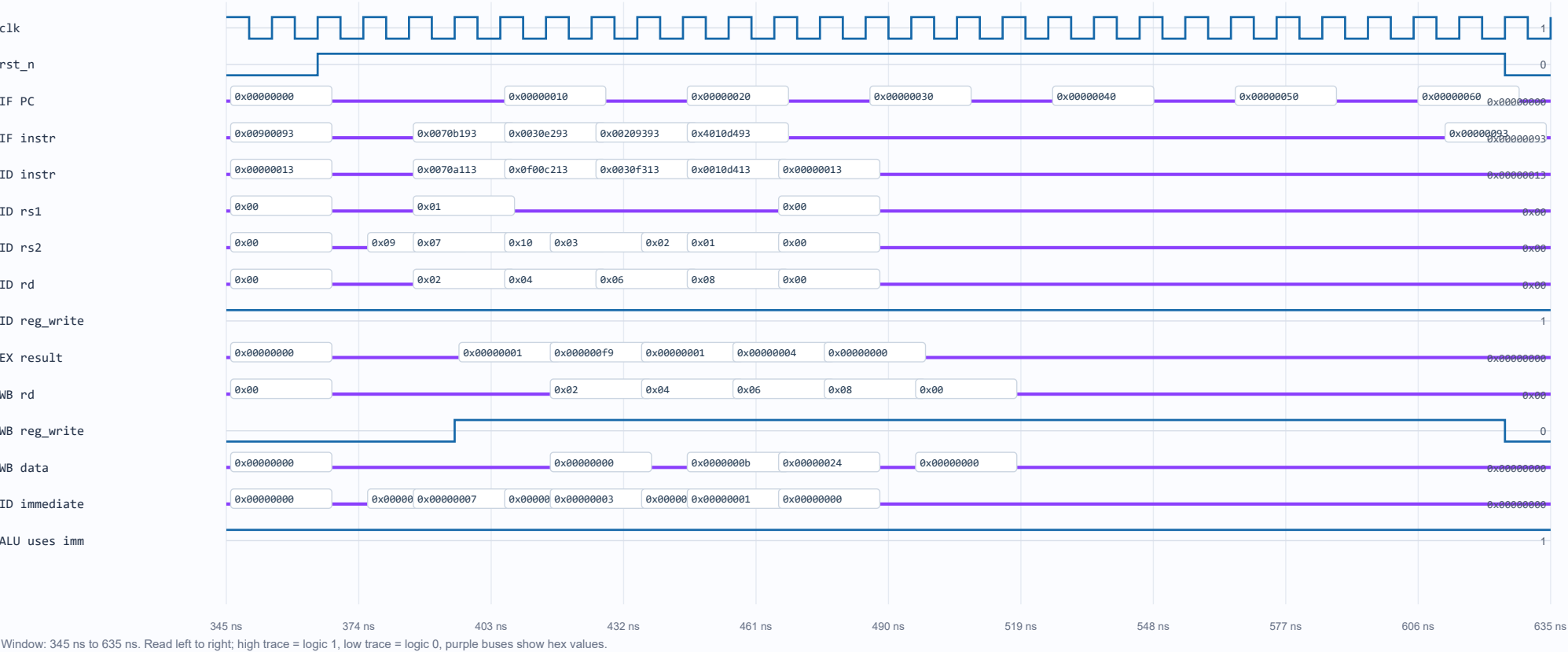
Beginner read: R-type instructions use two register operands. Watch rs1 and rs2 enter ID, EX result calculate, then WB data write into rd.



2. I-Type Instructions

ADDI, SLTI, SLTIU, XORI, ORI, ANDI, SLLI, SRLI, SRAI

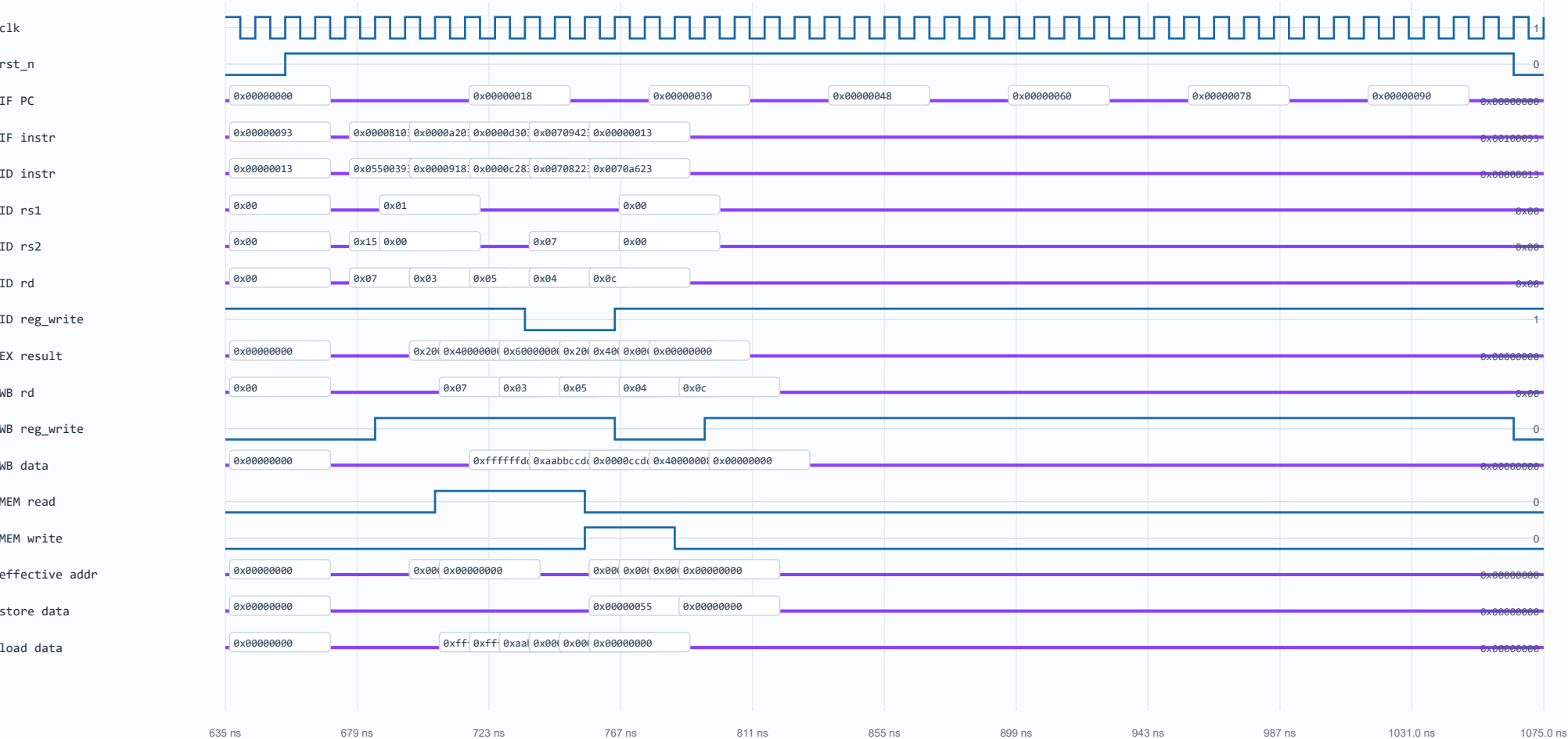
Beginner read: I-type instructions use rs1 plus an immediate. ALU uses imm goes high, the immediate appears in ID, and the result returns through WB.



3. Load and Store Instructions

LB, LH, LW, LBU, LHU, SB, SH, SW

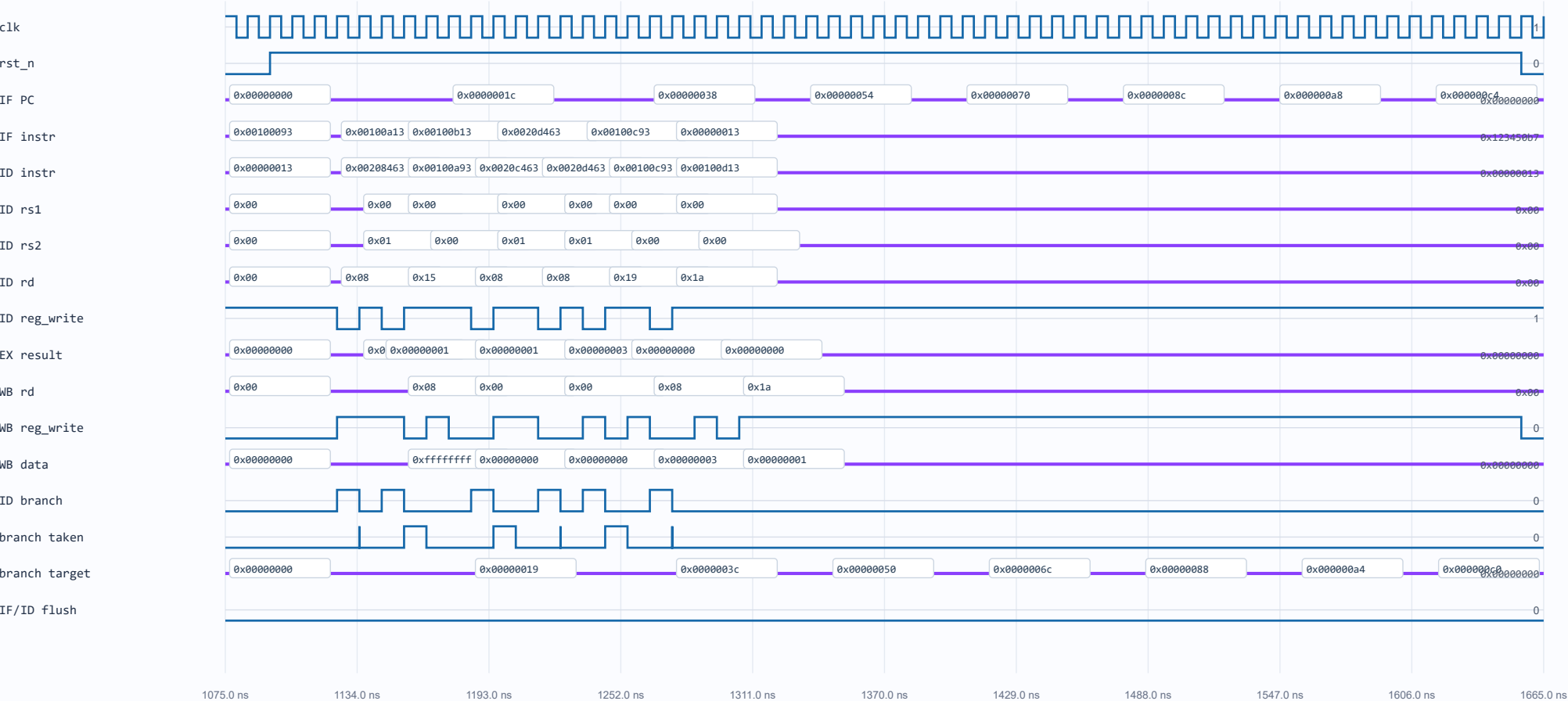
Beginner read: Loads assert MEM read and later write data back. Stores assert MEM write and use store data at the effective address.



4. Branch Instructions

BEQ, BNE, BLT, BGE, BLTU, BGEU

Beginner read: Branch decision happens in EX. If branch taken is high, the fetch/decode pipe is flushed and the PC redirects to branch target.

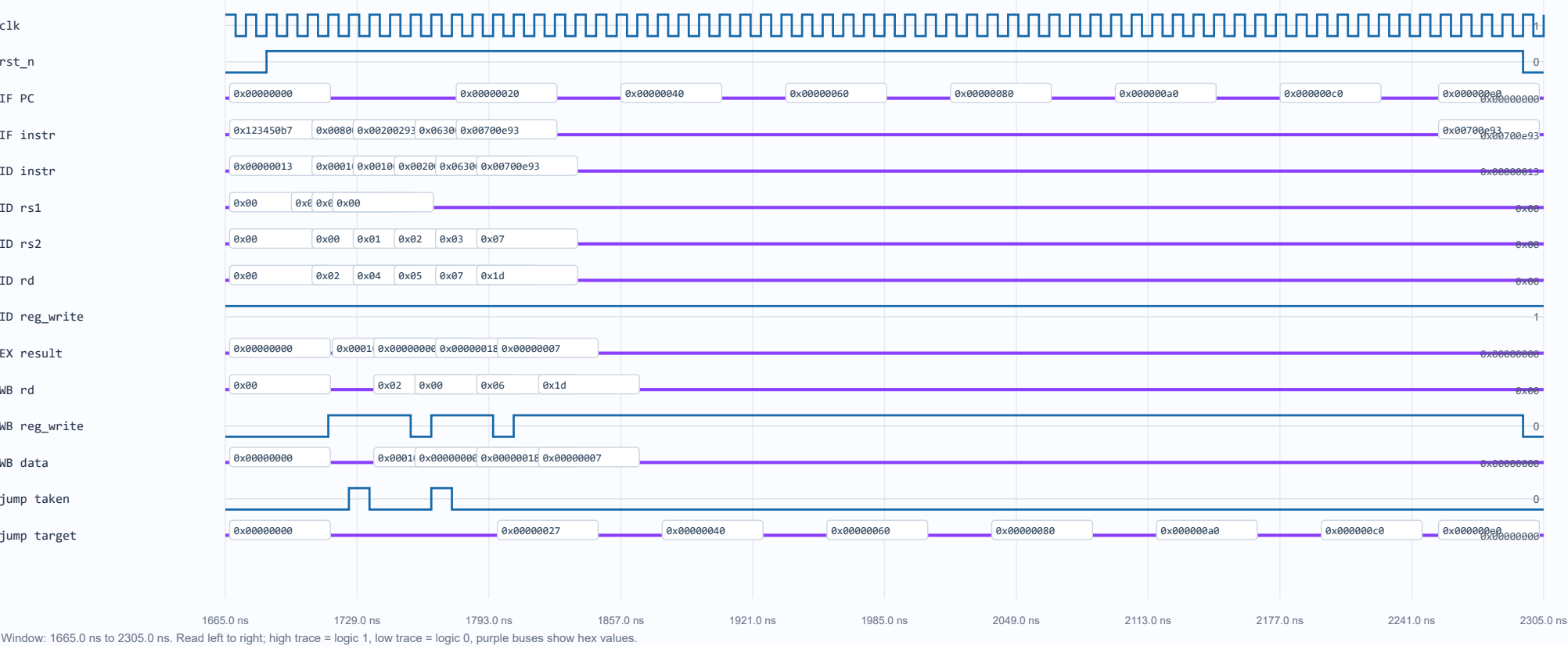


Window: 1075.0 ns to 1665.0 ns. Read left to right; high trace = logic 1, low trace = logic 0, purple buses show hex values.

5. U-Type and Jump Instructions

LUI, AUIPC, JAL, JALR

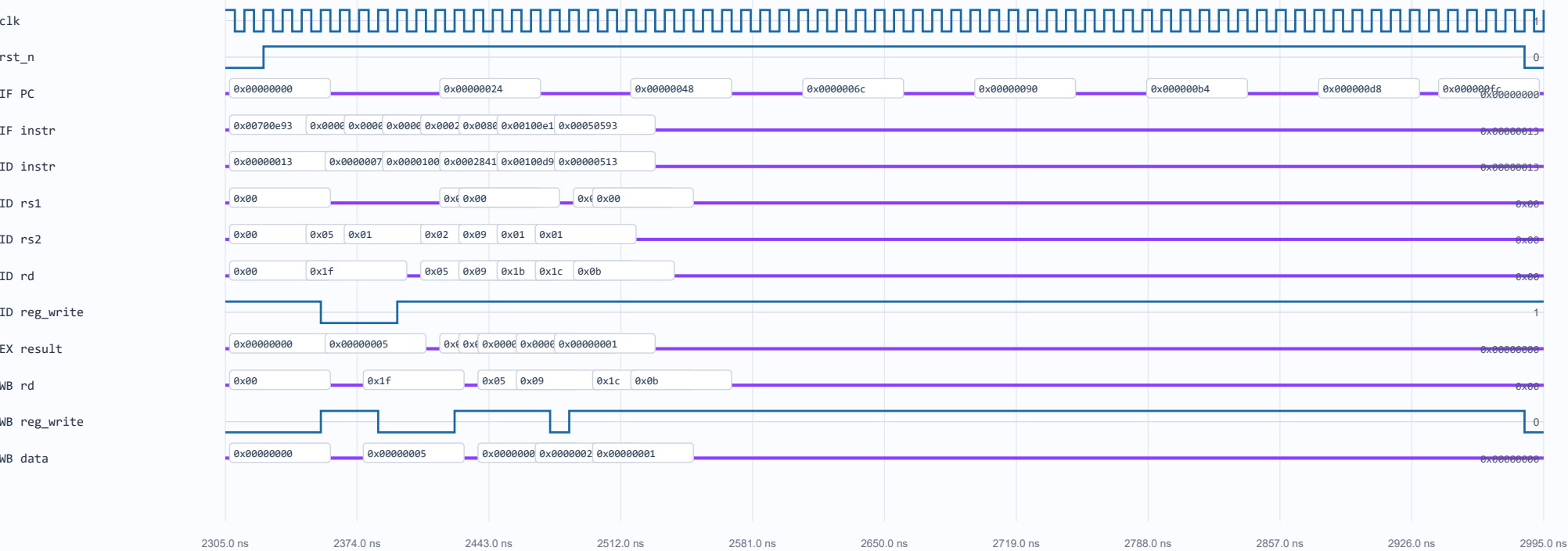
Beginner read: LUI/AUIPC create upper-immediate or PC-relative values. JAL/JALR redirect control flow and write PC+4 as the link value.



6. System, Fence, and Pseudo Instructions

ECALL, EBREAK, FENCE, FENCE.I, NOP, MV, LI, J

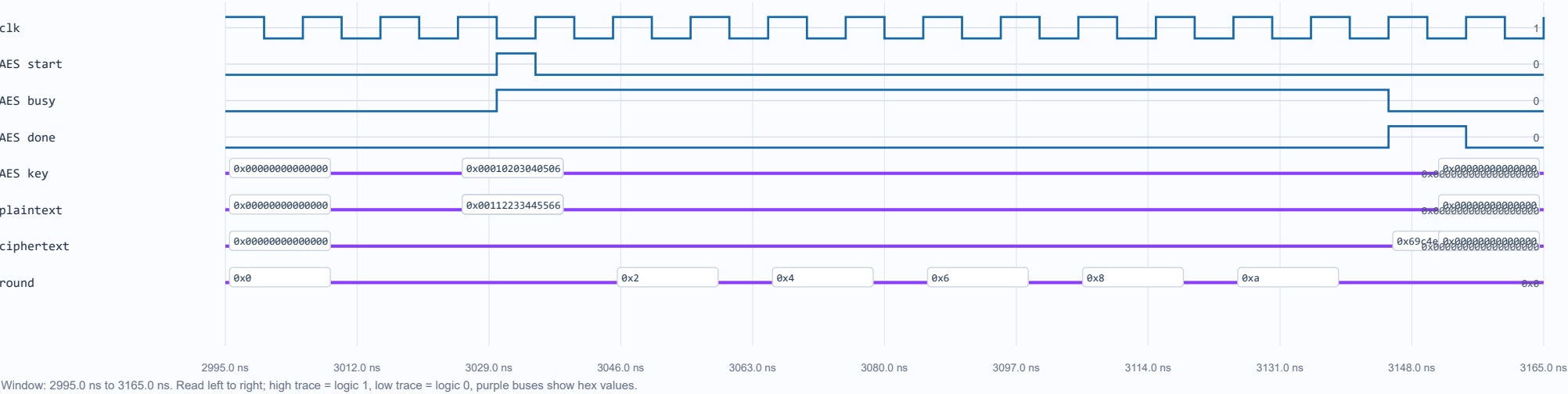
Beginner read: These tests show instructions that preserve state or map to ordinary RV32I operations. The important point is that WB only changes expected registers.



7. AES-128 ECB MMIO

NIST AES-128 known-answer vector

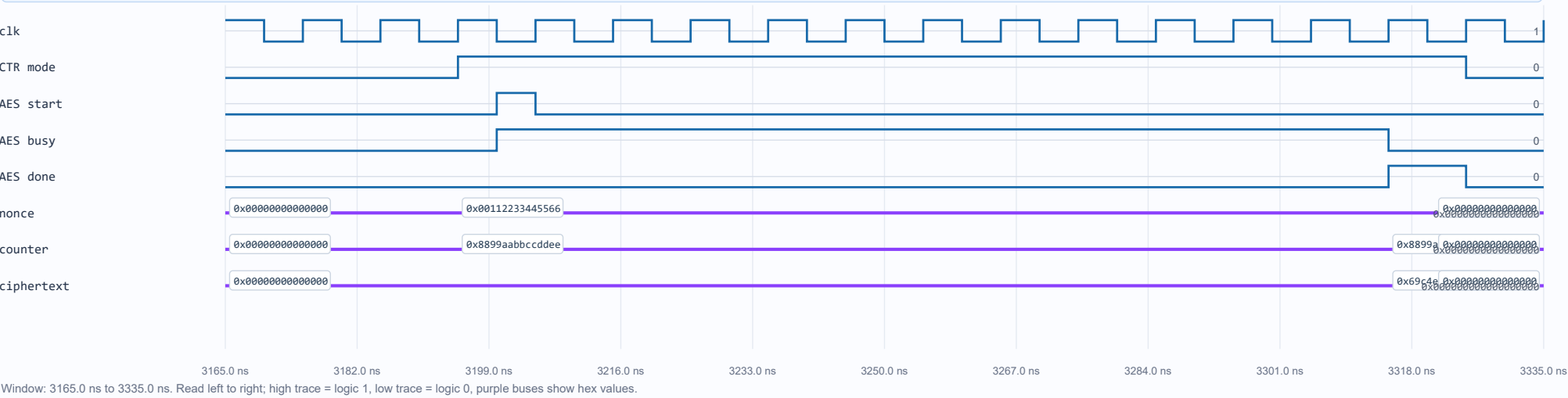
Beginner read: AES start pulses for one cycle, busy stays high while rounds execute, and done rises when ciphertext is valid.



8. AES-CTR Mode

AES keystream generation and counter increment

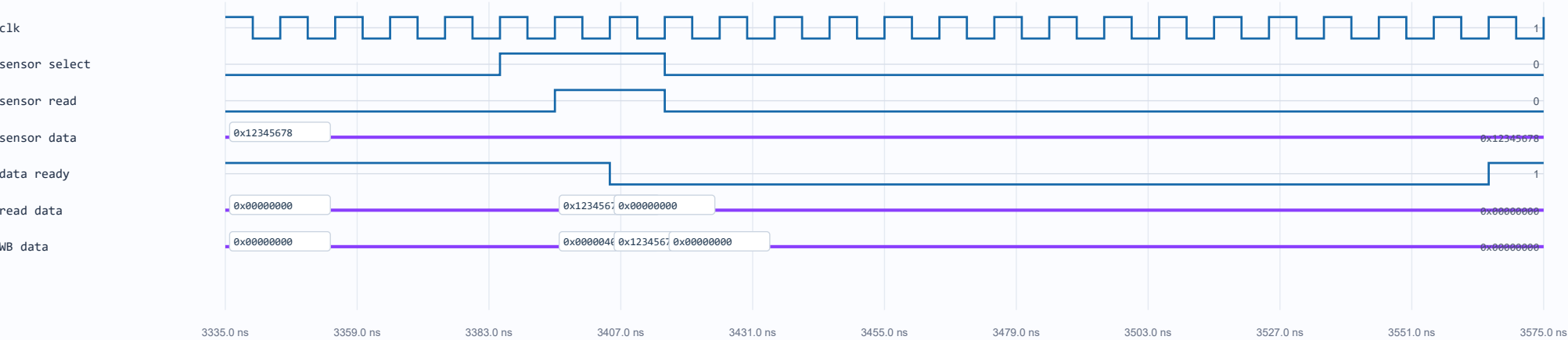
Beginner read: CTR mode encrypts nonce and counter to make a keystream. The counter increases after completion.



9. Sensor MMIO

Sensor data and status registers

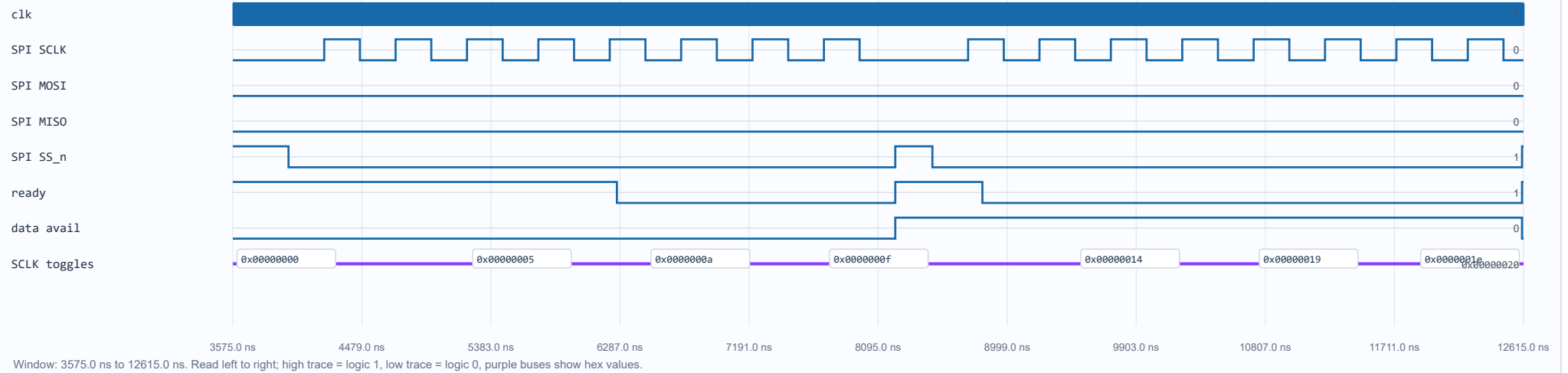
Beginner read: The CPU reads the sensor MMIO address. sensor select and sensor read show the access, then the value returns through WB.



10. Intel Avalon SPI IP

SPI master sensor interface

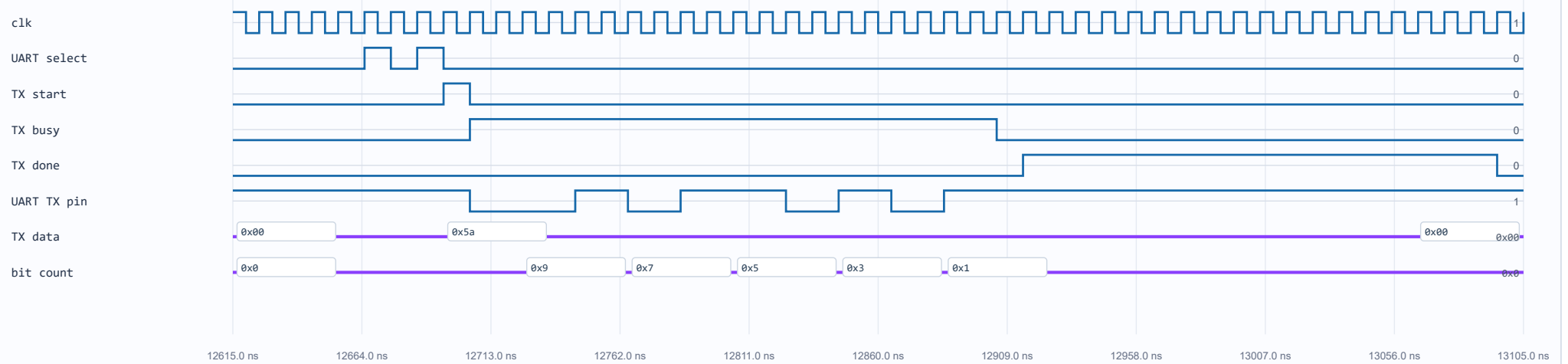
Beginner read: SCLK toggles during serial transfer. SS_n is active low, so a low value means the sensor slave is selected.



11. UART Transmit Peripheral

TX start, busy, done, and serial output

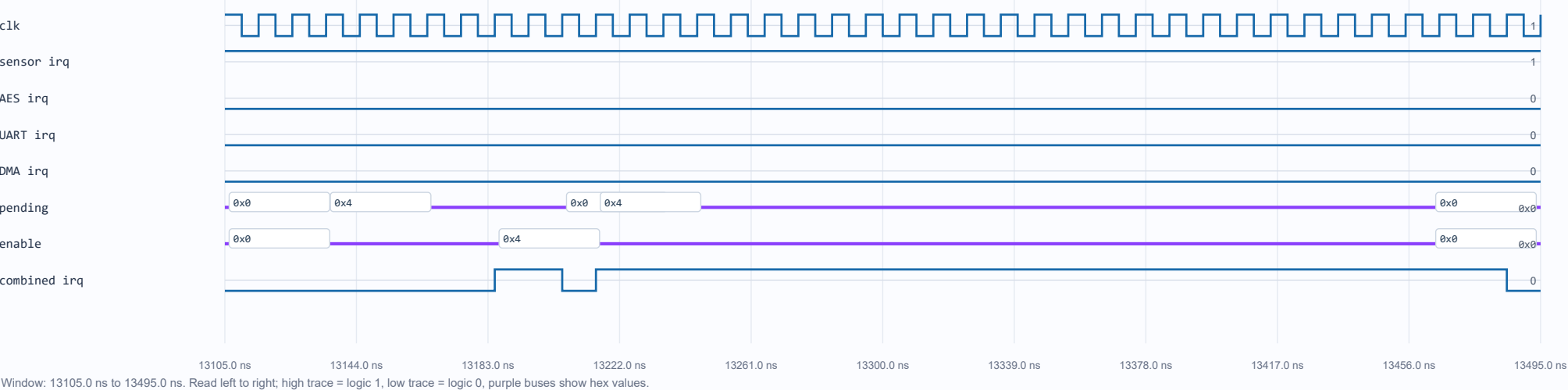
Beginner read: A TXDATA write creates TX start. busy stays high while serial bits shift on UART TX. done marks completion.



12. Interrupt Controller

Pending, enable, and combined IRQ

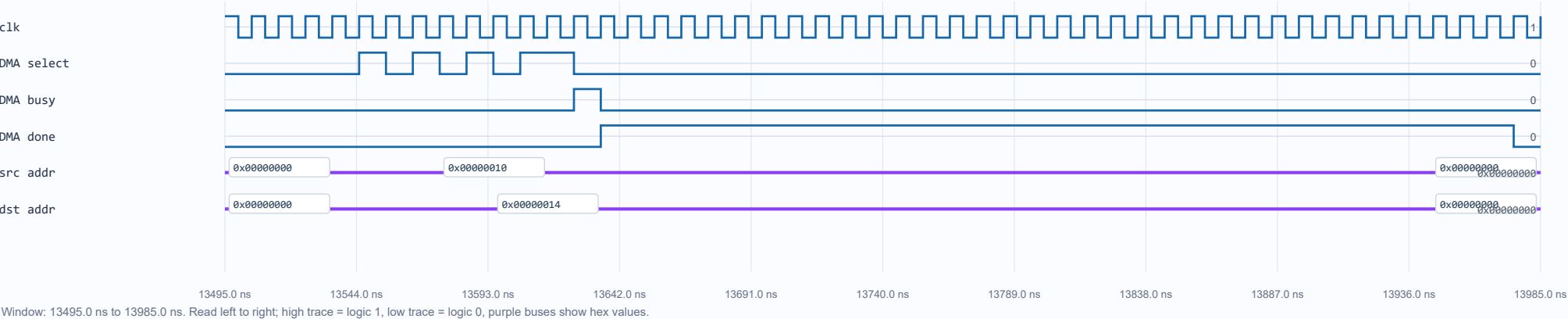
Beginner read: Source IRQs set pending bits. The enable register decides whether the combined IRQ line is asserted.



13. DMA-Lite Engine

Word transfer through internal memory side channel

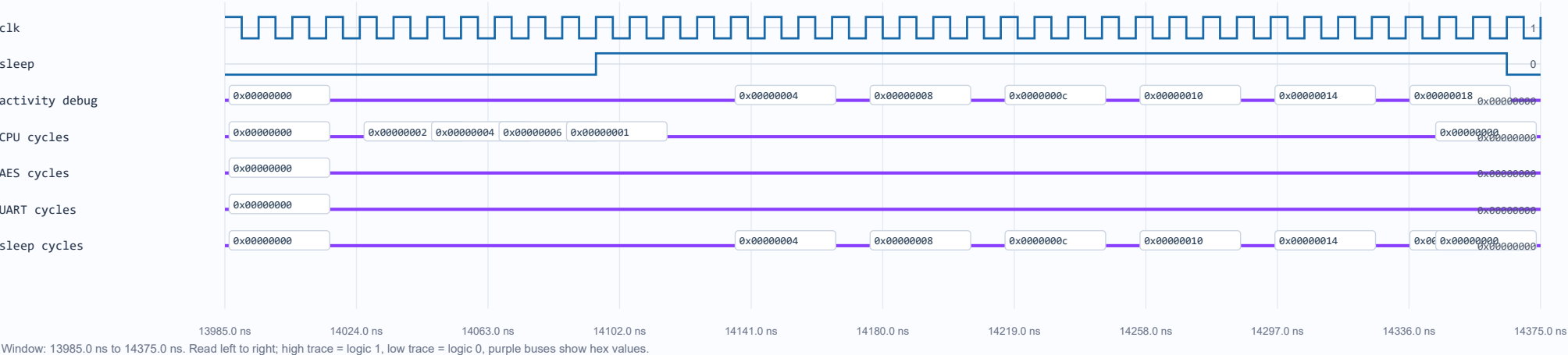
Beginner read: DMA copies a word from source to destination. busy shows transfer activity; done shows completion.



14. Power and Activity Counters

Sleep request and low-power measurements

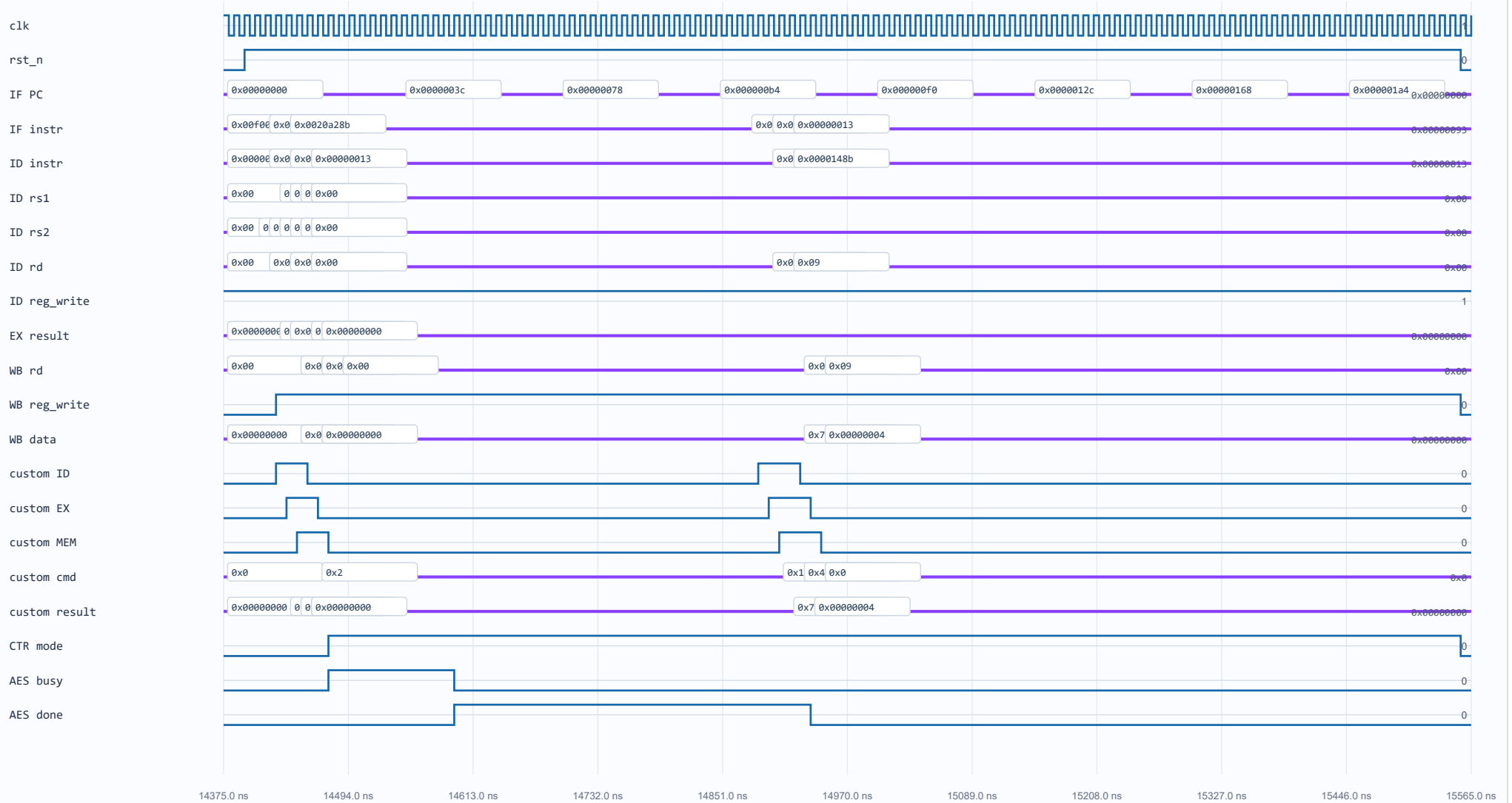
Beginner read: Activity counters increase when blocks are active. sleep indicates the low-power request and sleep cycles count idle time.



15. Custom Security ISA

RISC-V custom-0 AES/security instructions

Beginner read: custom ID, EX, and MEM show the custom instruction moving through the pipeline. custom cmd selects XOR, status, start, CT0 read, or clear.



Beginner Guide: How to Read These Waveforms

1. Time flows left to right

The bottom axis is simulation time. Each vertical grid line is a point in time.

2. Digital signals

A high line means logic 1. A low line means logic 0. Short pulses often mean one-cycle events.

3. Bus signals

Purple bus lines show multi-bit values such as PC, instruction, data, address, or ciphertext. Labels are hexadecimal.

4. Pipeline order

Instructions move IF to ID to EX to MEM to WB. Final register writes happen when WB reg_write is high.

5. MMIO peripherals

Peripheral select signals show which block is accessed: AES, sensor/SPI, UART, interrupt controller, DMA, or power block.

6. Custom ISA

custom ID, custom EX, and custom MEM show a custom instruction moving through the pipeline. custom cmd selects the security operation.